

Matrix Theory

1. Prove that for any matrices $A, B \in \mathbb{F}^{n \times n}$, $\text{tr}(AB) = \text{tr}(BA)$.
2. Use Gaussian Elimination to find the solution set for the linear system

$$\begin{array}{rcrcrcrcrcrl} x_1 & + & 2x_2 & - & 4x_3 & = & 10 \\ 2x_1 & - & x_2 & + & 2x_3 & = & 5 \\ x_1 & + & x_2 & - & 2x_3 & = & 7. \end{array}$$

Note. There are several methods you could use to solve this system, but use Gaussian Elimination to make sure you understand the algorithm!

